

UEFA Champions League 2021/22

EDA Project

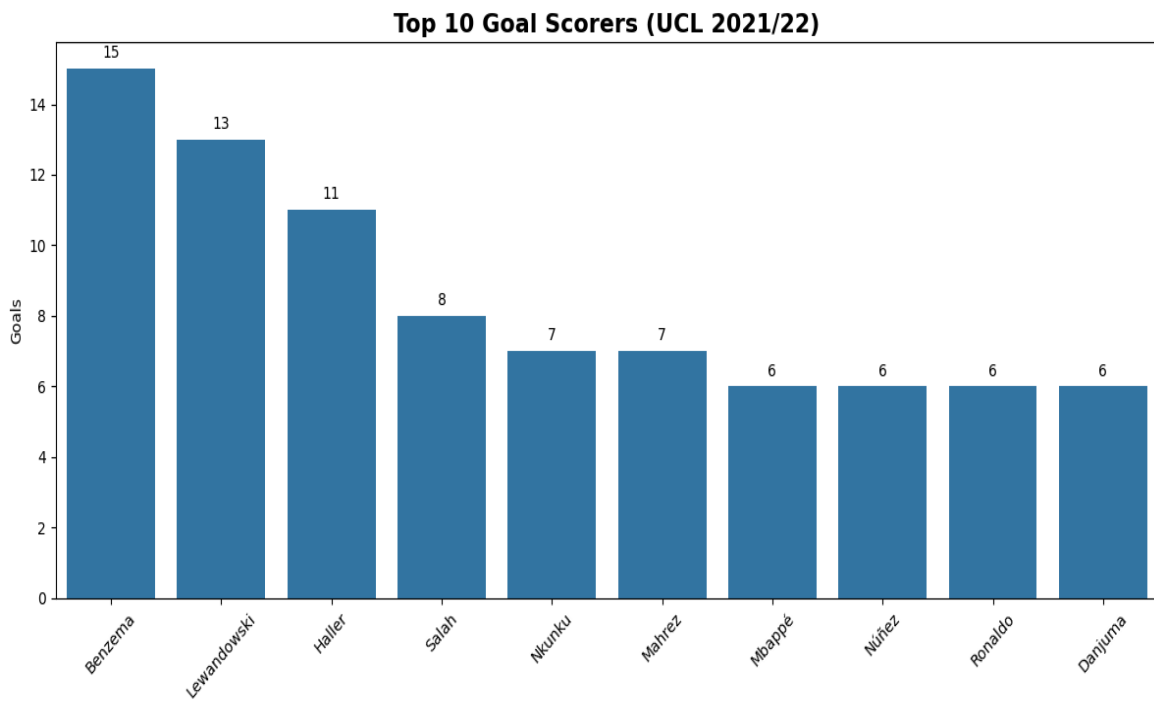
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Date-07/05/2026

Objectives-

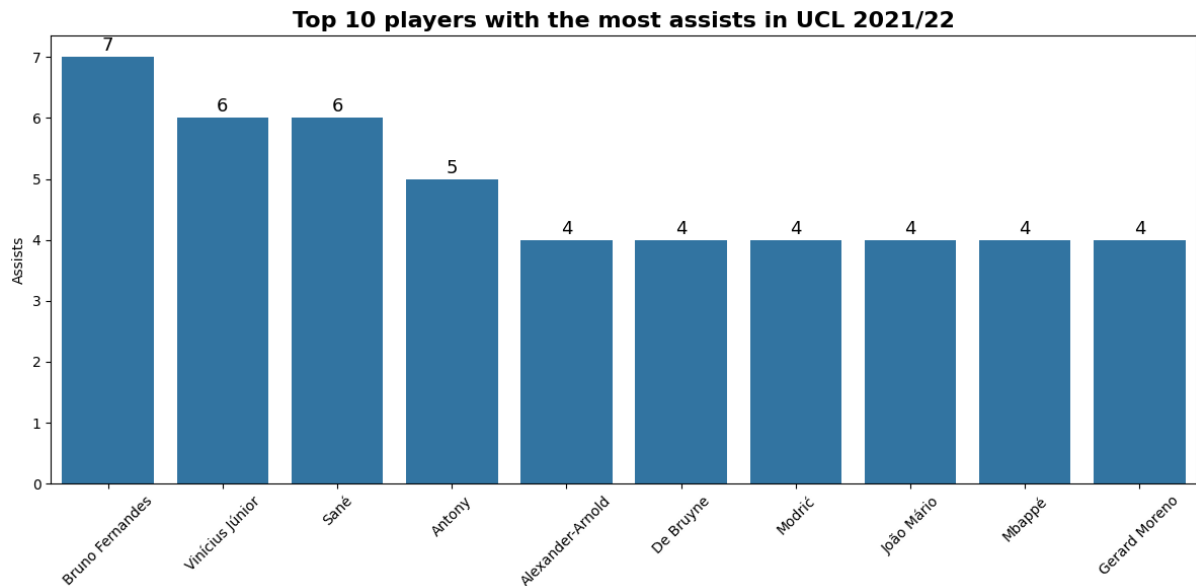
- To analyze player and team performance in the UEFA Champions League 2021/22 season
- To identify top performers based on goals, assists, and overall performance metrics
- To evaluate defenders and goalkeepers using composite scores
- To study relationships between variables such as goals, attempts, and assists
- To present insights using data visualization techniques

Graph 1 :- Top 10 Players with the most Goals
(UCL 2021/22)



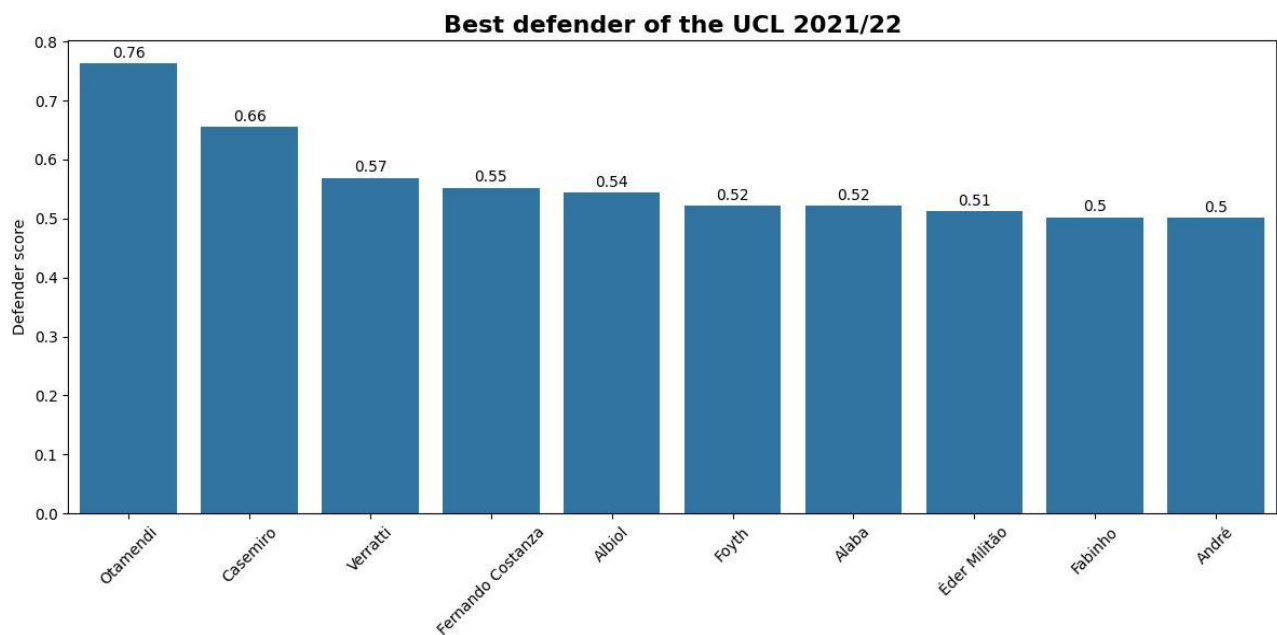
- The graph shows the top goal scorers of the UCL 2021/22 season, with Benzema leading the chart.
- A clear gap is seen between the top 3 players and the rest, indicating their dominance.

Graph 2 :- Top 10 Players with the most Assists (UCL 2021/22)



- The graph shows the top assist providers of the UCL 2021/22 season, with Bruno Fernandes leading the chart.
- Most players have similar assist numbers, indicating strong competition among playmakers.

Graph 3 :- The Best Defenders based on Defender Scores (UCL 2021/22)

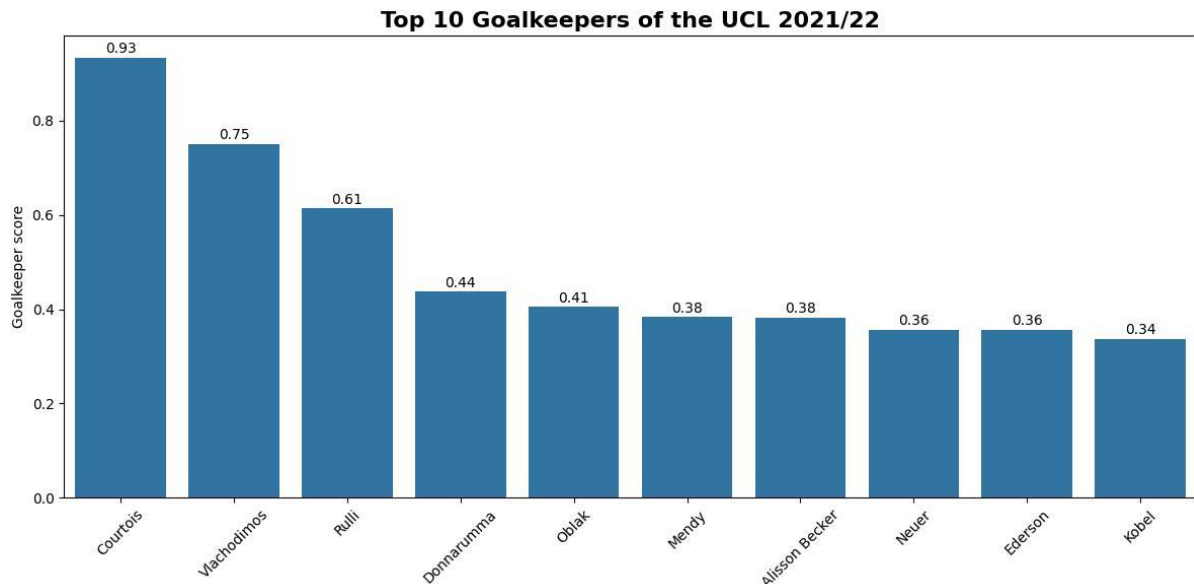


Methodology:

DEFENDER SCORE = $0.2 \times \text{TACKLES} + 0.3 \times \text{TACKLES WON} + 0.2 \times \text{CLEARANCES} + 0.2 \times \text{BALL RECOVERIES}$

- The graph ranks defenders based on a defender score, with Otamendi leading the chart.
- The score is calculated using tackles, tackles won, clearances, and ball recoveries, showing overall defensive performance.

Graph 4 :- The Best Goalkeepers based on Goalkeeper Score (UCL 2021/22)

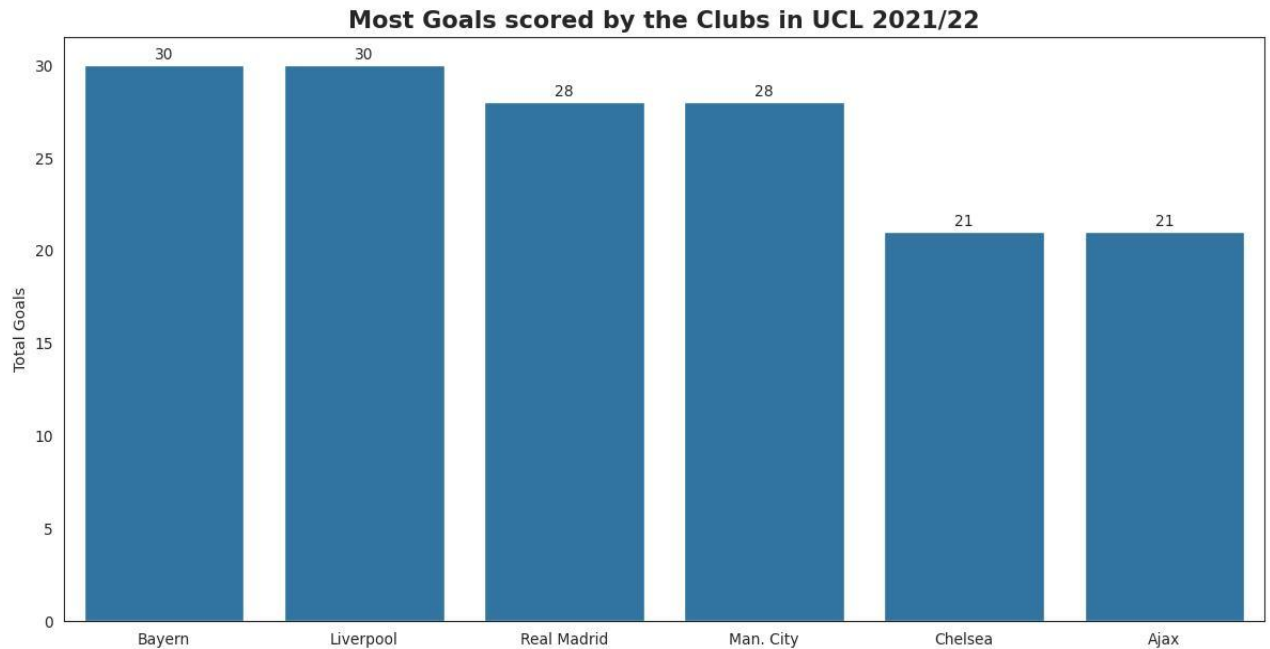


Methodology:

$$\text{GOALKEEPER SCORE} = 0.5 \times \text{SAVES} + 0.1 \times \text{PENALTY SAVES} + 0.3 \times \text{CLEAN SHEETS} + 0.1 \times \text{PUNCHES}$$

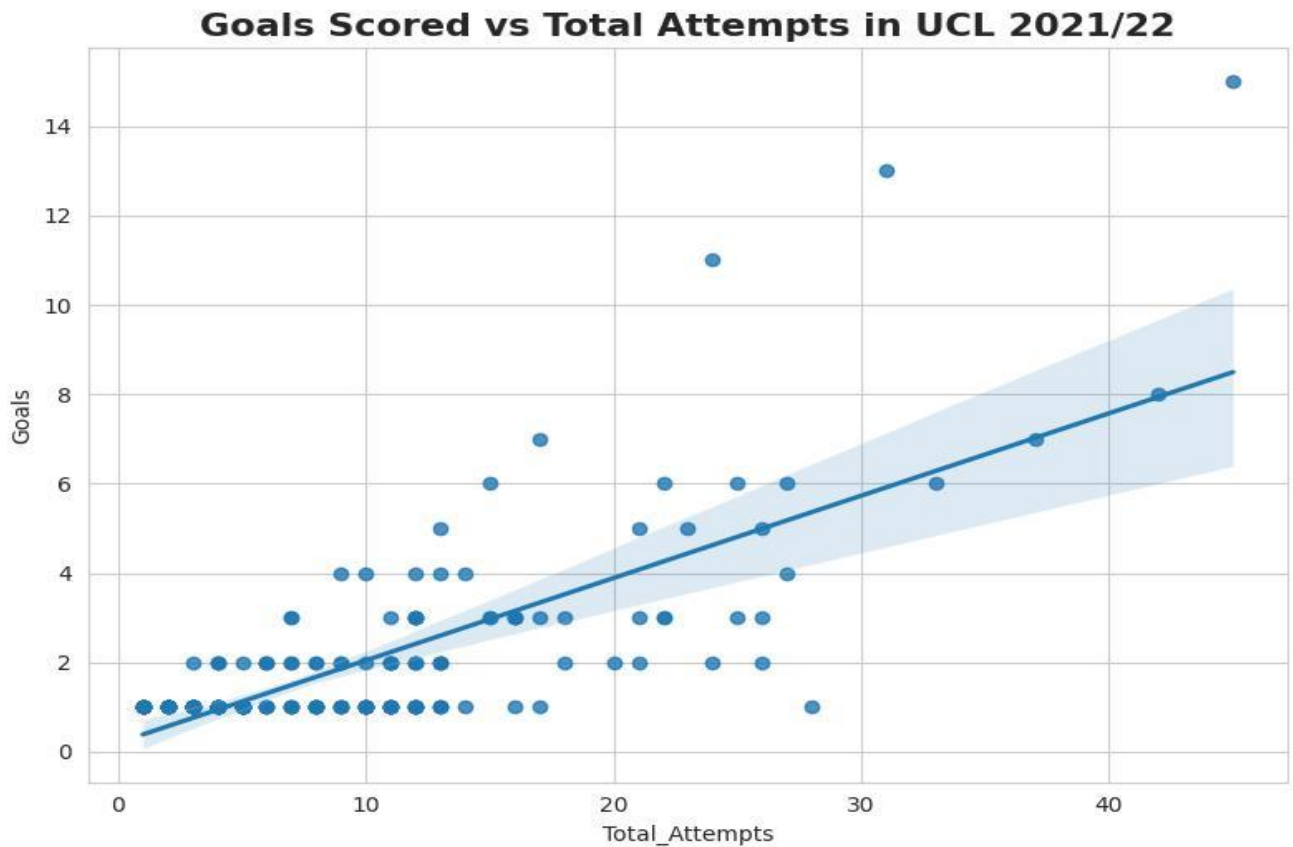
- The graph ranks goalkeepers based on a performance score, with Courtois leading the chart.
- The score combines saves, penalty saves, clean sheets, and punches to reflect overall goalkeeping performance.

**Graph 5 :- Football Clubs with the most Goals scored in
(UCL 2021/22)**



- The graph shows the total goals scored by different clubs, with Bayern and Liverpool leading the chart.
- Other top teams like Real Madrid and Manchester City are close behind, indicating strong attacking performance.

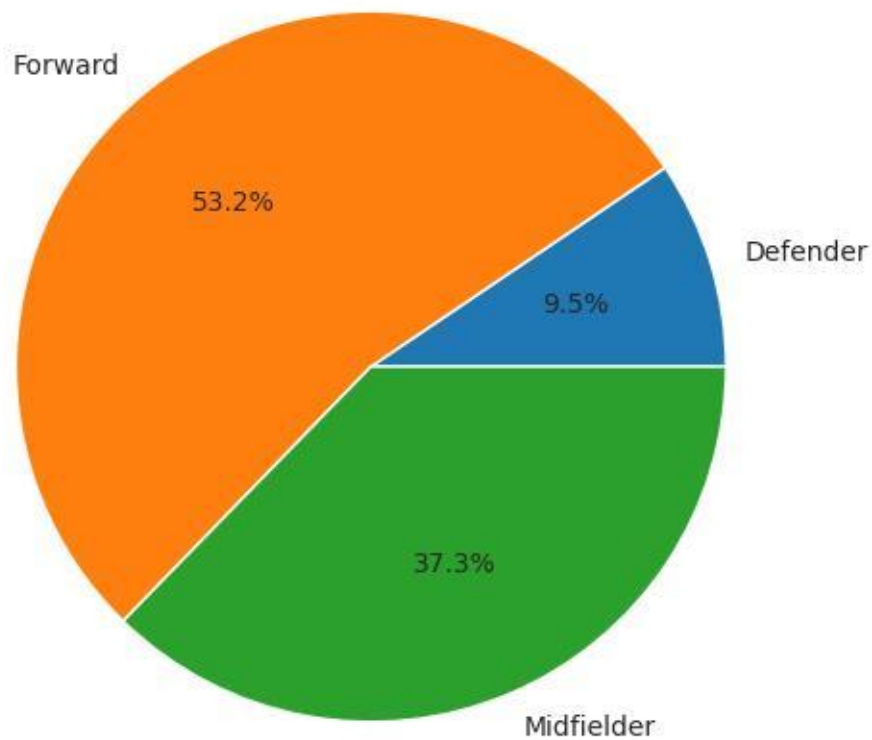
Graph 6 :- Goals Scored vs Total Attempts in
(UCL 2021/22)



- The graph shows the relationship between total attempts and goals scored, indicating a positive trend.
- Players who take more shots tend to score more goals, although efficiency varies among players.

Graph 7 :-Total Goals Scored by Positions in (UCL 2021/22)

Goal Distribution by Player Position (UCL 2021/22)



- The pie chart shows the distribution of goals by player position, with forwards contributing the majority.
- Midfielders contribute a moderate share, while defenders have minimal goal contribution.

Conclusion :-

This project is an exploratory data analysis (EDA) of the UEFA Champions League 2021/22 season, where I analyzed teams and players using different graphs and visualizations. Using bar plots, I was able to understand the top goal scorers, players with the highest assists, the best defenders using a defender score, and the best goalkeeper based on a goalkeeper score, along with the clubs that scored the most goals.

The regplot between goals and attempts showed that players who take more shots generally score more goals, although efficiency differs from player to player. The pie chart also showed that forwards score most of the goals compared to midfielders and defenders.

Overall, this EDA project helped me understand how data visualization makes it easier to analyze player performance and identify meaningful patterns in football data.

